Solutions to Judson's *Abstract Algebra*
Chapter 15: The Sylow Theorems, Exercises 1–26

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Preliminaries: tools used throughout

Let G be a finite group with $|G| = p^k m$, p prime, $p \nmid m$. A *Sylow p-subgroup* is a subgroup of order p^k, and n_p denotes their number. $N(H) = \{g \in G : gHg^{-1} = H\}$ is the normalizer and $C(a) = \{x : xa = ax\}$ the centralizer. A group is *simple* if its only normal subgroups are $\{\mathrm{e}\}$ and itself.

Theorem 1 (Sylow Theorems)**.** *(1) G has a Sylow p-subgroup (and more generally a subgroup of order p^i for each $p^i \mid |G|$).*
(2) Every p-subgroup of G lies in some Sylow p-subgroup, and any two Sylow p-subgroups are conjugate.
(3) $n_p \equiv 1 \pmod p$ and $n_p \mid m$.

Consequently a Sylow p-subgroup P is normal iff $n_p = 1$ (conjugates of P are again Sylow p-subgroups, and by (2) these are *all* of them).

Lemma 2 (Orbit counting of conjugates)**.** *For a subgroup $H \leq G$ the number of conjugates of H is $[G : N(H)]$. (This is Exercise 23 below.)*

Lemma 3 (p-groups have nontrivial center)**.** *If $|G| = p^r$ with $r \geq 1$, then $Z(G) \neq \{\mathrm{e}\}$. Hence G has a normal subgroup of order p.*

Proof. The class equation reads $|G| = |Z(G)| + \sum_i [G : C(x_i)]$, where the x_i are representatives of the conjugacy classes with more than one element. Each $[G : C(x_i)]$ is a divisor of p^r greater than 1, hence divisible by p. Since $p \mid |G|$, we get $p \mid |Z(G)|$, and so $|Z(G)| \geq p$. Pick $z \in Z(G)$, $z \neq \mathrm{e}$, of order p^j $(j \geq 1)$; then $w = z^{p^{j-1}}$ has order p and $\langle w \rangle$ is a normal subgroup (it is central) of order p. □

Lemma 4. *If $N \leq Z(G)$ and G/N is cyclic, then G is abelian.*

Proof. Let $G/N = \langle gN \rangle$. Every element of G has the form $g^i n$ with $n \in N$. For $x = g^i n$, $y = g^j n'$, since n, n' are central, $xy = g^{i+j} nn' = yx$. □

Lemma 5 (Kernel trick)**.** *Suppose G has $n > 1$ Sylow p-subgroups and $|G| > n!$. Then G has a normal subgroup other than $\{\mathrm{e}\}$ and G; in particular G is not simple.*

Proof. G acts on the set $\mathrm{Syl}_p(G)$ of its n Sylow p-subgroups by conjugation. By Theorem 1(2) this action is transitive, and since $n > 1$, some element moves some point. So the homomorphism $\varphi \colon G \to S_n$ is nontrivial and $K = \ker \varphi \neq G$. If $K = \{\mathrm{e}\}$, then φ would be injective, forcing $|G| \leq n!$, contrary to the hypothesis. Thus K is a proper nontrivial normal subgroup. □

Lemma 6. *Every group of order p^2 (p prime) is abelian, hence isomorphic to $\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.*

Proof. By Lemma 3, $|Z(G)| \in \{p, p^2\}$. If $|Z(G)| = p$, then $G/Z(G)$ has order p so is cyclic, and G is abelian by Lemma 4, contradicting $Z(G) \neq G$. So $Z(G) = G$. The second statement follows from the Fundamental Theorem of Finite Abelian Groups. □

Exercises

Exercise 1. *What are the orders of all Sylow p-subgroups where G has order 18, 24, 54, 72, and 80?*

Solution. A Sylow p-subgroup has order equal to the largest power of p dividing $|G|$.

$\lvert G\rvert$	Factorization	Sylow p-subgroup orders
18	$2 \cdot 3^2$	$p=2$: 2; $p=3$: 9
24	$2^3 \cdot 3$	$p=2$: 8; $p=3$: 3
54	$2 \cdot 3^3$	$p=2$: 2; $p=3$: 27
72	$2^3 \cdot 3^2$	$p=2$: 8; $p=3$: 9
80	$2^4 \cdot 5$	$p=2$: 16; $p=5$: 5

■

Exercise 2. *Find all the Sylow 3-subgroups of S_4 and show that they are all conjugate.*

Solution. $|S_4| = 24 = 2^3 \cdot 3$, so Sylow 3-subgroups have order 3, i.e. they are generated by elements of order 3. The elements of order 3 in S_4 are the eight 3-cycles, and each subgroup of order 3 contains exactly two of them (c and c^{-1}). So there are exactly four Sylow 3-subgroups:

$$P_4 = \{(1), (123), (132)\}, \qquad P_3 = \{(1), (124), (142)\},$$
$$P_2 = \{(1), (134), (143)\}, \qquad P_1 = \{(1), (234), (243)\}.$$

(Consistent with Sylow: $4 \equiv 1 \pmod 3$ and $4 \mid 8$.)

Conjugacy. For $\sigma \in S_4$ we have $\sigma(a\,b\,c)\sigma^{-1} = (\sigma(a)\ \sigma(b)\ \sigma(c))$, hence $\sigma\langle(a\,b\,c)\rangle\sigma^{-1} = \langle(\sigma(a)\ \sigma(b)\ \sigma(c))\rangle$. Given any two of the subgroups, say $\langle(a\,b\,c)\rangle$ and $\langle(a'\,b'\,c')\rangle$, choose $\sigma \in S_4$ with $\sigma(a) = a'$, $\sigma(b) = b'$, $\sigma(c) = c'$ (and σ sending the remaining point to the remaining point). Then $\sigma\langle(a\,b\,c)\rangle\sigma^{-1} = \langle(a'\,b'\,c')\rangle$. So all four are conjugate. ■

Exercise 3. *Show that every group of order 45 has a normal subgroup of order 9.*

Solution. $45 = 3^2 \cdot 5$. A Sylow 3-subgroup has order 9. By Theorem 1(3), $n_3 \mid 5$ and $n_3 \equiv 1 \pmod 3$, so $n_3 \in \{1, 5\}$. Since $5 \equiv 2 \pmod 3$, $n_3 = 1$. Hence the unique Sylow 3-subgroup is normal, and it has order 9. ■

Exercise 4. *Let H be a Sylow p-subgroup of G. Prove that H is the only Sylow p-subgroup of G contained in $N(H)$.*

Solution. Let $|G| = p^k m$ with $p \nmid m$, so $|H| = p^k$. Clearly $H \leq N(H)$, and $H \trianglelefteq N(H)$ by definition of $N(H)$.

Let K be a Sylow p-subgroup of G with $K \leq N(H)$. Then $|K| = p^k$. Since p^k is the largest power of p dividing $|G|$, it is also the largest power of p dividing $|N(H)|$ (a divisor of $|G|$). So H and K are both Sylow p-subgroups of the group $N(H)$. By Sylow's theorem applied in $N(H)$, there is $x \in N(H)$ with $K = xHx^{-1}$. But H is normal in $N(H)$, so $xHx^{-1} = H$. Therefore $K = H$. ■

Exercise 5. *Prove that no group of order* 96 *is simple.*

Solution. $96 = 2^5 \cdot 3$. By Theorem 1(3), $n_2 \mid 3$, so $n_2 \in \{1, 3\}$.

- If $n_2 = 1$, the Sylow 2-subgroup (of order 32) is a proper nontrivial normal subgroup.
- If $n_2 = 3$, then $|G| = 96 > 3! = 6$, so Lemma 5 gives a proper nontrivial normal subgroup.

In either case G is not simple. ■

Exercise 6. *Prove that no group of order* 160 *is simple.*

Solution. $160 = 2^5 \cdot 5$. By Theorem 1(3), $n_2 \mid 5$, so $n_2 \in \{1, 5\}$.

- If $n_2 = 1$, the Sylow 2-subgroup (order 32) is a proper nontrivial normal subgroup.
- If $n_2 = 5$, then $|G| = 160 > 5! = 120$, so Lemma 5 gives a proper nontrivial normal subgroup.

Hence G is not simple. ■

Exercise 7. *If H is a normal subgroup of a finite group G and $|H| = p^k$ for some prime p, show that H is contained in every Sylow p-subgroup of G.*

Solution. Let P be any Sylow p-subgroup of G. Since $H \trianglelefteq G$, the product HP is a subgroup of G, and

$$|HP| = \frac{|H|\,|P|}{|H \cap P|},$$

which is a power of p (as $|H|$ and $|P|$ are). Also $|HP|$ divides $|G|$, so $|HP| \leq p^{k'}$ where $p^{k'} = |P|$ is the largest power of p dividing $|G|$. But $P \leq HP$ gives $|HP| \geq |P|$. Hence $|HP| = |P|$, and $P \leq HP$ forces $HP = P$. Therefore $H \leq HP = P$. ■

Exercise 8. *Let G be a group of order p^2q^2, where p and q are distinct primes such that $q \nmid p^2 - 1$ and $p \nmid q^2 - 1$. Prove that G must be abelian. List three pairs of primes satisfying these conditions.*

Solution. **Sylow p-subgroups.** By Theorem 1(3), $n_p \mid q^2$ and $n_p \equiv 1 \pmod{p}$, so $n_p \in \{1, q, q^2\}$.

- If $n_p = q$, then $p \mid q - 1$, so $p \mid (q-1)(q+1) = q^2 - 1$, contradicting $p \nmid q^2 - 1$.
- If $n_p = q^2$, then $p \mid q^2 - 1$, again a contradiction.

So $n_p = 1$, and the Sylow p-subgroup P (order p^2) is normal.

Sylow q-subgroups. Symmetrically $n_q \in \{1, p, p^2\}$ with $n_q \equiv 1 \pmod{q}$. If $n_q = p$ then $q \mid p - 1 \mid p^2 - 1$; if $n_q = p^2$ then $q \mid p^2 - 1$; both contradict $q \nmid p^2 - 1$. So $n_q = 1$ and the Sylow q-subgroup Q (order q^2) is normal.

Conclusion. $P, Q \trianglelefteq G$ and $P \cap Q = \{\mathrm{e}\}$ (their orders are coprime), so for $x \in P$, $y \in Q$ the commutator $xyx^{-1}y^{-1}$ lies in both P and Q, hence equals e. Also $|PQ| = p^2q^2 = |G|$, so $G \cong P \times Q$. By Lemma 6, P and Q are abelian, therefore G is abelian.

Examples. Neither prime can be 2 or 3: if $p = 2$ then $2 \mid q^2 - 1$ for every odd prime q, and if $p = 3$ then $3 \mid q^2 - 1$ for every prime $q \neq 3$. For primes ≥ 5 we check:

(p, q)	$p^2 - 1$	$q^2 - 1$	conditions
$(5, 7)$	24	48	$7 \nmid 24$, $5 \nmid 48$
$(5, 13)$	24	168	$13 \nmid 24$, $5 \nmid 168$
$(7, 11)$	48	120	$11 \nmid 48$, $7 \nmid 120$

So $(5,7)$, $(5,13)$, $(7,11)$ work (as does, e.g., $(11,13)$). ■

Exercise 9. *Show directly that a group of order 33 has only one Sylow 3-subgroup.*

Solution. $33 = 3 \cdot 11$. By Theorem 1(3), $n_3 \mid 11$ and $n_3 \equiv 1 \pmod 3$, so $n_3 \in \{1, 11\}$. Since $11 \equiv 2 \pmod 3$, $n_3 \neq 11$. Therefore $n_3 = 1$. ■

Exercise 10. *Let H be a subgroup of a group G. Prove or disprove that the normalizer of H is normal in G.*

Solution. **The statement is false.** Take $G = S_3$ and $H = \langle (12) \rangle = \{(1), (12)\}$. Then $N(H) = \{g : g(12)g^{-1} \in H\} = \{g : g(12)g^{-1} = (12)\} = C((12))$. The conjugates $(13)(12)(13)^{-1} = (23)$ and $(123)(12)(132) = (23)$ are different from (12), so $N(H) = \{(1), (12)\} = H$. But H is not normal in S_3: $(123)(12)(123)^{-1} = (23) \notin H$. So $N(H)$ is not normal in G. ■

Exercise 11. *Let G be a finite group divisible by a prime p. Prove that if there is only one Sylow p-subgroup in G, it must be a normal subgroup of G.*

Solution. Let P be the unique Sylow p-subgroup. For any $g \in G$, gPg^{-1} is a subgroup of the same order $|P|$, hence also a Sylow p-subgroup. By uniqueness, $gPg^{-1} = P$ for all $g \in G$, so $P \trianglelefteq G$. ■

Exercise 12. *Let G be a group of order p^r, p prime. Prove that G contains a normal subgroup of order p^{r-1}.*

Solution. We assume $r \geq 1$ and use induction on r.

Base case $r = 1$. The trivial subgroup $\{\mathrm{e}\}$ has order p^0 and is normal.

Inductive step. Let $r \geq 2$ and assume the claim for groups of order p^{r-1}. By Lemma 3, G has a normal subgroup N of order p. The quotient G/N has order p^{r-1}, so by the induction hypothesis it has a normal subgroup $\overline{M}$ of order p^{r-2}. Let $\pi \colon G \to G/N$ be the quotient map and $M = \pi^{-1}(\overline{M})$. As the preimage of a normal subgroup under a homomorphism, $M \trianglelefteq G$. The restriction $\pi|_M \colon M \to \overline{M}$ is onto with kernel N, so $|M| = |N| \cdot |\overline{M}| = p \cdot p^{r-2} = p^{r-1}$. ■

Exercise 13. *Suppose that G is a finite group of order $p^n k$, where $k < p$. Show that G must contain a normal subgroup.*

Solution. Assume $n \geq 1$. Since $k < p$, we have $p \nmid k$, so p^n is the full power of p in $|G|$ and $m = k$. By Theorem 1(3), $n_p \mid k$ and $n_p \equiv 1 \pmod p$. Suppose $n_p > 1$. Then $n_p \equiv 1 \pmod p$ gives $n_p \geq p + 1$, while $n_p \mid k$ gives $n_p \leq k < p$, a contradiction. So $n_p = 1$, and the Sylow p-subgroup, of order p^n, is normal in G.

Remark. This normal subgroup is nontrivial; it is proper when $k > 1$. If $k = 1$, then G itself is a p-group and by Exercise 12 it has a normal subgroup of order p^{n-1}, which is proper and nontrivial for $n \geq 2$ (when $n = 1$, $G \cong \mathbb{Z}_p$ and there is nothing more to say).

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Exercise 14. *Let H be a subgroup of a finite group G. Prove that $gN(H)g^{-1} = N(gHg^{-1})$ for any $g \in G$.*

Solution. ($\subseteq$) Let $x \in N(H)$. Then

$$(gxg^{-1})(gHg^{-1})(gxg^{-1})^{-1} = g\,(xHx^{-1})\,g^{-1} = gHg^{-1},$$

so $gxg^{-1} \in N(gHg^{-1})$.

($\supseteq$) Let $y \in N(gHg^{-1})$ and put $x = g^{-1}yg$. Then

$$xHx^{-1} = g^{-1}y\,(gHg^{-1})\,y^{-1}g = g^{-1}(gHg^{-1})g = H,$$

so $x \in N(H)$ and $y = gxg^{-1} \in gN(H)g^{-1}$. ■

Exercise 15. *Prove that a group of order* 108 *must have a normal subgroup.*

Solution. $108 = 2^2 \cdot 3^3$. By Theorem 1(3), $n_3 \mid 4$ and $n_3 \equiv 1 \pmod 3$, so $n_3 \in \{1, 4\}$.

- If $n_3 = 1$, the Sylow 3-subgroup (order 27) is normal.
- If $n_3 = 4$, then $|G| = 108 > 4! = 24$, so Lemma 5 gives a proper nontrivial normal subgroup.

■

Exercise 16. *Classify all the groups of order* 175 *up to isomorphism.*

Solution. $175 = 5^2 \cdot 7$.

- $n_5 \mid 7$ and $n_5 \equiv 1 \pmod 5$: $n_5 \in \{1, 7\}$ and $7 \equiv 2 \pmod 5$, so $n_5 = 1$.
- $n_7 \mid 25$ and $n_7 \equiv 1 \pmod 7$: $n_7 \in \{1, 5, 25\}$ with $5 \not\equiv 1$ and $25 \equiv 4 \pmod 7$, so $n_7 = 1$.

Thus the Sylow 5-subgroup P (order 25) and Sylow 7-subgroup Q (order 7) are both normal. As in Exercise 8, $P \cap Q = \{\mathrm{e}\}$ and $|PQ| = 175$ give $G \cong P \times Q$. Now $Q \cong \mathbb{Z}_7$ and, by Lemma 6, $P \cong \mathbb{Z}_{25}$ or $\mathbb{Z}_5 \times \mathbb{Z}_5$. Therefore, up to isomorphism, there are exactly two groups of order 175 (both abelian):

$$\mathbb{Z}_{25} \times \mathbb{Z}_7 \cong \mathbb{Z}_{175}, \qquad \mathbb{Z}_5 \times \mathbb{Z}_5 \times \mathbb{Z}_7 \cong \mathbb{Z}_5 \times \mathbb{Z}_{35}.$$

■

Exercise 17. *Show that every group of order* 255 *is cyclic.*

Solution. $255 = 3 \cdot 5 \cdot 17$.

Step 1: groups of order 15 *are cyclic.* If $|K| = 15$, then $n_3 \mid 5$, $n_3 \equiv 1 \pmod 3$ gives $n_3 = 1$ (as $5 \equiv 2$), and $n_5 \mid 3$, $n_5 \equiv 1 \pmod 5$ gives $n_5 = 1$. Both Sylow subgroups are normal with trivial intersection, so $K \cong \mathbb{Z}_3 \times \mathbb{Z}_5 \cong \mathbb{Z}_{15}$.

Step 2: the Sylow 17*-subgroup is normal.* $n_{17} \mid 15$ and $n_{17} \equiv 1 \pmod{17}$. The divisors of 15 are $1, 3, 5, 15$, and only 1 is $\equiv 1 \pmod{17}$. So $n_{17} = 1$, and the Sylow 17-subgroup $P \cong \mathbb{Z}_{17}$ is normal.

Step 3: P is central. G acts on P by conjugation, giving a homomorphism $\psi\colon G \to \mathrm{Aut}(P) \cong \mathrm{Aut}(\mathbb{Z}_{17})$, which has order $\varphi(17) = 16$. The order of $\psi(G) \cong G/\ker\psi$ divides both $|G| = 255$ and 16; since $\gcd(255, 16) = 1$, $\psi(G)$ is trivial. So every $g \in G$ acts trivially: $gxg^{-1} = x$ for all $x \in P$, i.e. $P \le Z(G)$.

Step 4. G/P has order 15, so it is cyclic by Step 1. Since $P \le Z(G)$, Lemma 4 shows G is abelian. An abelian group of order $255 = 3 \cdot 5 \cdot 17$ (distinct primes) is $\cong \mathbb{Z}_3 \times \mathbb{Z}_5 \times \mathbb{Z}_{17} \cong \mathbb{Z}_{255}$. Hence G is cyclic. ■

Exercise 18. *Let G have order $p_1^{e_1} \cdots p_n^{e_n}$ (p_i distinct primes) and suppose that G has n Sylow p-subgroups $P_1, \ldots, P_n$ where $|P_i| = p_i^{e_i}$. Prove that G is isomorphic to $P_1 \times \cdots \times P_n$.*

Solution. We read the hypothesis as: for each prime p_i there is exactly one Sylow p_i-subgroup P_i (so G has exactly n Sylow subgroups in total). By Exercise 11 each $P_i \trianglelefteq G$.

Elements of different P_i commute. Let $i \neq j$, $x \in P_i$, $y \in P_j$. Since P_i, P_j are normal, $xyx^{-1}y^{-1} = (xyx^{-1})y^{-1} \in P_j$ and also $= x(yx^{-1}y^{-1}) \in P_i$. But $P_i \cap P_j = \{\mathrm{e}\}$ because $\gcd(|P_i|, |P_j|) = 1$. So $xy = yx$.

The map. Define $\varphi\colon P_1 \times \cdots \times P_n \to G$, $\varphi(x_1, \ldots, x_n) = x_1 x_2 \cdots x_n$. Because elements of different factors commute,

$$\varphi\big((x_i)(y_i)\big) = \prod_i x_i y_i = \Big(\prod_i x_i\Big)\Big(\prod_i y_i\Big),$$

so φ is a homomorphism.

Injective. Suppose $x_1 \cdots x_n = \mathrm{e}$. Fix i. Then $x_i^{-1} = \prod_{j \neq i} x_j$. The order of x_i divides $p_i^{e_i}$. On the other hand the elements x_j ($j \neq i$) commute pairwise and have orders dividing $p_j^{e_j}$, so the order of $\prod_{j\neq i} x_j$ divides $\prod_{j \neq i} p_j^{e_j}$. These two numbers are coprime, so $x_i = \mathrm{e}$. Thus $\ker \varphi$ is trivial.

Onto. $|P_1 \times \cdots \times P_n| = \prod p_i^{e_i} = |G|$, so the injective map φ is a bijection. Hence $G \cong P_1 \times \cdots \times P_n$. ■

Exercise 19. *Let P be a normal Sylow p-subgroup of G. Prove that every inner automorphism of G fixes P.*

Solution. Here "fixes P" means that each inner automorphism $\iota_g(x) = gxg^{-1}$ maps P onto itself (as a set): $\iota_g(P) = gPg^{-1} = P$. This is exactly the statement that P is normal, which is given.

For completeness: $\iota_g(P) = gPg^{-1} \subseteq P$ by normality, and applying this to g^{-1} gives $g^{-1}Pg \subseteq P$, i.e. $P \subseteq gPg^{-1}$; so $\iota_g(P) = P$ and ι_g restricts to an automorphism of P.

(Moreover, *every* automorphism α of G satisfies $\alpha(P) = P$: $\alpha(P)$ is a subgroup of order $|P|$, hence a Sylow p-subgroup, and P is the only one since it is normal.)

Note that this does *not* mean the automorphisms fix each element of P; e.g. in S_3 the normal Sylow 3-subgroup A_3 is mapped to itself by conjugation by (12), which swaps (123) and (132). ■

Exercise 20. *What is the smallest possible order of a group G such that G is nonabelian and $|G|$ is odd? Can you find such a group?*

Solution. **The answer is 21.**

Odd orders below 21 give abelian groups. The odd orders are $1, 3, 5, 7, 9, 11, 13, 15, 17, 19$. Groups of prime order are cyclic, hence abelian; groups of order $9 = 3^2$ are abelian by Lemma 6; groups of order 15 are cyclic by Step 1 of Exercise 17. So any nonabelian group of odd order has order at least 21.

A nonabelian group of order 21. Let G be the set of affine maps of $\mathbb{Z}_7$,

$$G = \{\, f_{u,c}\colon x \mapsto ux + c \;:\; u \in \{1, 2, 4\},\ c \in \mathbb{Z}_7 \,\}.$$

Note $\{1, 2, 4\}$ is a subgroup of $\mathbb{Z}_7^\times$ (it is $\{1, 2, 2^2\}$ with $2^3 = 8 \equiv 1$). Composition satisfies $f_{u,c} \circ f_{u',c'} = f_{uu',\, uc'+c}$, which lies in G, and $f_{u,c}^{-1} = f_{u^{-1},\, -u^{-1}c} \in G$. So G is a subgroup of the group of bijections of $\mathbb{Z}_7$, and $|G| = 3 \cdot 7 = 21$. It is nonabelian: for $f(x) = x + 1$ and $g(x) = 2x$,

$$(f \circ g)(x) = 2x + 1 \neq 2x + 2 = (g \circ f)(x).$$

Equivalently, $G = \langle a, b \mid a^7 = b^3 = \mathrm{e},\ bab^{-1} = a^2 \rangle \cong \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ (here $a = f_{1,1}$, $b = f_{2,0}$), or as a permutation group, $G = \langle (1234567), (235)(476) \rangle \leq S_7$. ■

Exercise 21. ***The Frattini Lemma.*** *If H is a normal subgroup of a finite group G and P is a Sylow p-subgroup of H, for each $g \in G$ show that there is an $h \in H$ such that $gPg^{-1} = hPh^{-1}$. Also, show that if N is the normalizer of P, then $G = HN$.*

Solution. Let $g \in G$. Since $H \trianglelefteq G$, $gPg^{-1} \leq gHg^{-1} = H$, and $|gPg^{-1}| = |P|$. So gPg^{-1} is a subgroup of H of the same order as the Sylow p-subgroup P of H; hence it is also a Sylow p-subgroup of H. By Sylow's theorem applied in H, the Sylow p-subgroups of H are conjugate *in* H, so there is $h \in H$ with

$$gPg^{-1} = hPh^{-1}.$$

Then $(h^{-1}g)P(h^{-1}g)^{-1} = h^{-1}(gPg^{-1})h = h^{-1}(hPh^{-1})h = P$, so $n := h^{-1}g \in N = N(P)$. Therefore $g = hn \in HN$. Since g was arbitrary, $G = HN$ (the inclusion $HN \subseteq G$ is trivial). ■

Exercise 22. *Show that if the order of G is $p^n q$, where p and q are primes and $p > q$, then G contains a normal subgroup.*

Solution. By Theorem 1(3), $n_p \mid q$ and $n_p \equiv 1 \pmod{p}$, so $n_p \in \{1, q\}$. If $n_p = q$, then $p \mid q - 1$; but $0 < q - 1 < p$, which is impossible. Hence $n_p = 1$ and the Sylow p-subgroup (of order p^n) is normal in G. (It is nontrivial for $n \geq 1$ and proper since $q > 1$.) This is the special case $k = q < p$ of Exercise 13. ■

Exercise 23. *Prove that the number of distinct conjugates of a subgroup H of a finite group G is $[G : N(H)]$.*

Solution. Let X be the (finite) set of subgroups of G. G acts on X by conjugation, $g \cdot K = gKg^{-1}$: indeed $\mathrm{e}K\mathrm{e}^{-1} = K$ and $(g_1g_2)K(g_1g_2)^{-1} = g_1(g_2Kg_2^{-1})g_1^{-1}$. The orbit of H is the set of conjugates $\{gHg^{-1}\}$ of H, and its stabilizer is

$$G_H = \{g : gHg^{-1} = H\} = N(H).$$

By the Orbit–Stabilizer theorem (Judson Thm. 14.11), the number of conjugates is $|\mathcal{O}_H| = [G : N(H)]$. ■

Exercise 24. *Prove that a Sylow 2-subgroup of S_5 is isomorphic to D_4.*

Solution. $|S_5| = 120 = 2^3 \cdot 15$, so Sylow 2-subgroups have order 8. Consider the symmetry group of the square with vertices $1, 2, 3, 4$ in order, regarded as a subgroup of S_5 fixing 5:

$$D = \{(1), (1234), (13)(24), (1432), (13), (24), (12)(34), (14)(23)\}.$$

(The rotations are the powers of (1234); the reflections are the two diagonal reflections $(13), (24)$ and the two edge reflections $(12)(34), (14)(23)$.) This is a subgroup of $S_4 \leq S_5$ isomorphic to D_4 and of order 8, so it is a Sylow 2-subgroup of S_5. By Sylow's theorem every Sylow 2-subgroup of S_5 is conjugate to D, hence isomorphic to $D \cong D_4$. ■

Exercise 25. ***Another Proof of the Sylow Theorems.*** *Let $|G| = p^k m$ with $p \nmid m$. (a) Show that $p \nmid \binom{p^k m}{p^k}$. (b) Let S denote the set of all p^k-element subsets of G. Show that $p \nmid |S|$. (c) Define an action of G on S by left multiplication, $aT = \{at : t \in T\}$ for $a \in G$, $T \in S$. Prove that this is a group action. (d) Prove $p \nmid |\mathcal{O}_T|$ for some $T \in S$. (e) Let $\{T_1, \ldots, T_u\}$ be an orbit such that $p \nmid u$ and $H = \{g \in G : gT_1 = T_1\}$. Prove that H is a subgroup of G and show that $|G| = u|H|$. (f) Show that p^k divides $|H|$ and $p^k \leq |H|$. (g) Show that $|H| = |\mathcal{O}_t| \leq p^k$ for $t \in T_1$ (orbit under H); conclude that $p^k = |H|$.*

Solution. Put $N = p^k m$ and $r = p^k$.

(a) We have $\binom{N}{r} = \dfrac{N}{r}\binom{N-1}{r-1} = m\prod_{i=1}^{r-1}\dfrac{N-i}{r-i}$. For $1 \le i \le r-1$ write $i = p^j u$ with $p \nmid u$ and $0 \le j < k$. Then

$$N - i = p^j\big(p^{k-j}m - u\big), \qquad r - i = p^j\big(p^{k-j} - u\big),$$

and since $k - j \ge 1$ and $p \nmid u$, neither $p^{k-j}m - u$ nor $p^{k-j} - u$ is divisible by p. So each fraction $\frac{N-i}{r-i} = \frac{p^{k-j}m-u}{p^{k-j}-u}$ has numerator and denominator prime to p. Therefore the integer $\binom{N}{r}$ equals m times a ratio of integers prime to p, and $p \nmid m$; hence $p \nmid \binom{N}{r}$.

(b) $|S| = \binom{p^k m}{p^k}$, which is not divisible by p by (a).

(c) For $a \in G$ and $T \in S$, $|aT| = |T| = p^k$ (left multiplication by a is injective), so $aT \in S$. Further $eT = T$ and $(ab)T = \{abt\} = a\{bt : t \in T\} = a(bT)$. So this is a group action.

(d) S is the disjoint union of the orbits, so $|S| = \sum |\mathcal{O}_{T_i}|$ over orbit representatives. If p divided every orbit size, p would divide $|S|$, contradicting (b). So some orbit $\mathcal{O}_T$ has size not divisible by p.

(e) $H = \{g : gT_1 = T_1\}$ is the stabilizer of T_1, hence a subgroup (Judson Prop. 14.10). By the Orbit–Stabilizer theorem, $u = |\mathcal{O}_{T_1}| = [G : H]$, i.e. $|G| = u|H|$.

(f) $p^k \mid |G| = u|H|$ and $p \nmid u$, so $\gcd(p^k, u) = 1$ and hence $p^k \mid |H|$. In particular $p^k \le |H|$.

(g) Fix $t \in T_1$. Then H acts on T_1 by left multiplication (as $hT_1 = T_1$), and the orbit of t is $Ht = \{ht : h \in H\} \subseteq T_1$. The map $H \to Ht$, $h \mapsto ht$, is a bijection (if $ht = h't$ then $h = h'$), so $|H| = |Ht| \le |T_1| = p^k$. Combined with (f), $|H| = p^k$.

Thus H is a subgroup of G of order p^k: a Sylow p-subgroup exists. This reproves the first Sylow theorem. ∎

Exercise 26. *Let G be a group. Prove that $G' = \langle aba^{-1}b^{-1} : a, b \in G\rangle$ is a normal subgroup of G and G/G' is abelian. Find an example to show that $\{aba^{-1}b^{-1} : a, b \in G\}$ is not necessarily a group.*

Solution. Write $[a, b] = aba^{-1}b^{-1}$ and let $\mathcal{C} = \{[a, b] : a, b \in G\}$, so $G' = \langle\mathcal{C}\rangle$.

G' is normal. For $g \in G$,

$$g[a, b]g^{-1} = (gag^{-1})(gbg^{-1})(ga^{-1}g^{-1})(gb^{-1}g^{-1}) = [gag^{-1}, gbg^{-1}] \in \mathcal{C},$$

so conjugation by g maps $\mathcal{C}$ into $\mathcal{C}$. Since $x \mapsto gxg^{-1}$ is a homomorphism, it maps the subgroup generated by $\mathcal{C}$ into the subgroup generated by $\mathcal{C}$: $gG'g^{-1} \subseteq G'$ for every g. Hence $G' \trianglelefteq G$.

G/G' is abelian. For $a, b \in G$,

$$(aG')(bG')(aG')^{-1}(bG')^{-1} = aba^{-1}b^{-1}G' = G',$$

since $[a, b] \in G'$. So $(aG')(bG') = (bG')(aG')$, and G/G' is abelian.

The set $\mathcal{C}$ of commutators need not be a subgroup. Let $V = \mathbb{F}_2^4$ with standard basis $e_1, \dots, e_4$, and let $W = \Lambda^2 V$ be the $\mathbb{F}_2$-vector space with basis $e_{ij} = e_i \wedge e_j$ $(1 \le i < j \le 4)$, so $\dim W = 6$. For $v, w \in V$ put

$$v \wedge w = \sum_{i<j} p_{ij}(v, w)\, e_{ij}, \qquad p_{ij}(v, w) = v_i w_j - v_j w_i,$$

and define the bilinear map $\beta\colon V\times V\to W$, $\beta(v,w)=\sum_{i<j} v_i w_j\, e_{ij}$. Then $\beta(v,w)-\beta(w,v)=v\wedge w$. Let

$$G = V\times W, \qquad (v,x)(v',x') = \big(v+v',\ x+x'+\beta(v,v')\big).$$

Since β is bilinear, this operation is associative (both ways of bracketing give $x+x'+x''+\beta(v,v')+\beta(v,v'')+\beta(v',v'')$), $(0,0)$ is the identity, and $(v,x)^{-1}=(-v,-x+\beta(v,v))$. So G is a group of order $2^4\cdot 2^6=1024$. The elements $(0,x)$ are central.

Commutators. For $s=(v,x)$, $s'=(v',x')$ we have $ss'=(v+v',x+x'+\beta(v,v'))$ and $s's=(v+v',x+x'+\beta(v',v))$, so $ss'=z\,s's$ with $z=(0,\beta(v,v')-\beta(v',v))=(0,v\wedge v')$ central. Hence

$$[s,s'] = ss's^{-1}s'^{-1} = z\,(s's)s^{-1}s'^{-1} = z = (0,\ v\wedge v').$$

So $\mathcal{C}=\{(0,\ v\wedge w) : v,w\in V\}$: the commutators are exactly the "decomposable" elements $v\wedge w$ of W.

G' contains a non-commutator. Each basis element $e_{ij}=e_i\wedge e_j$ is decomposable, so $(0,e_{ij})\in\mathcal{C}\subseteq G'$, and therefore $(0,x)\in G'$ for every $x\in W$. In particular $g_0=(0,\ e_{12}+e_{34})\in G'$. Suppose $g_0\in\mathcal{C}$, so $e_{12}+e_{34}=v\wedge w$ for some $v,w\in V$, i.e. $p_{12}=p_{34}=1$ and $p_{13}=p_{14}=p_{23}=p_{24}=0$ (where $p_{ij}=p_{ij}(v,w)$). Expanding the polynomials shows the identity

$$p_{12}p_{34}-p_{13}p_{24}+p_{14}p_{23}=0$$

for all v,w (the six monomials $v_iv_jw_kw_l$ cancel in pairs). But for our values the left-hand side equals $1\cdot 1-0+0=1\neq 0$ in $\mathbb{F}_2$, a contradiction. So $g_0\notin\mathcal{C}$.

Thus g_0 is a product of two commutators, $g_0=(0,e_{12})(0,e_{34})$, but not itself a commutator: $\mathcal{C}$ is not closed under multiplication, so it is not a subgroup of G, while $G'=\{(0,x) : x\in W\}$ is.

Remark. Smaller examples exist (the smallest group in which the set of commutators is not a subgroup has order 96), but this one can be verified completely by hand.

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